

EDUCATION

Indraprastha Institute of Information Technology, Delhi
BTech in Computer Science and Applied Mathematics; CGPA- 8.68/10

Delhi, India
2019 - 2023

WORK EXPERIENCE

Atlassian *[Software company that develops collaboration tools (Jira & Confluence) for developers and PMs]*

Bengaluru

Associate Data Scientist

Jun 2023 – Present

- Collaborated with product managers to perform Geo-location/Cloud-usage cohort analysis that helped reduce cloud residency cost for 200+ Enterprise customers by ~ 8% , at the same time decreasing their page response time by 15%
- Highlighted opportunities to drive Cloud Trial adoption based on the effectiveness of past promos and a sales signal-based On-Prem to Cloud Migration propensity model; Painted a customer journey view using Plotly and Tableau for the Demand Generation Team to understand customer behaviour and build custom nurture emails for customers, leading to additional 25k+ trial signups
- Built an optimized usage-based pricing and packaging strategy for a new storage backup and restore product, using linear regression for projecting customer adoption and costs, saving > \$1.5k /customer/month in unused-storage costs for Atlassian.
- Instrumented a novel metric that measures the dollar-impact of On-Prem to Cloud Migrations to help leadership better evaluate the performance of a 4-year long migration strategy decision; used by the finance team in our Quarterly Earnings Release to showcase the impact of company's cloud-migration efforts
- Facilitated the development and launch of top requested features in Atlassian's cloud migration assistant software through quantitative survey analysis and A/B Testing over multiple sprints, also led the development of 6 reliability and performance metrics for our cloud migration software suite
- Delivered 10+ deep-dive analyses and Key Dashboards in support of Level (1/2/3) OKRs to senior leadership, and positively influenced cloud migration efforts across Sales Team, Channel partners and Finance Dept
- Established a process-driven source control framework for all Data Science ETLs used by the team, successfully migrated 50+ legacy ETLs to the new Databricks, Bitbucket, and Apache Spark architecture, making subsequent collaborations efficient; Built a Data-Quality Recommendation system using Anthropic's Claude 3.5 fine tuned on internal SQL tables to suggest fundamental DQ checks and save 10+Hr/Week of developer time.

Atlassian

Remote

Data Science Intern

May – Aug 2022

- Developed a recommendation system using a Random-Forest classifier that suggests the best on-prem to cloud migration strategy tailored for each customer with a MAP@[k=3] score of 0.70, saving hours of effort and time for our Enterprise Partners.
- Engineered a pipeline using PySpark & Plotly that analyzes post migration-to-cloud survey responses, enriches the data from multiple sources, identifies key drivers and generates insights & visualizations on a clean dashboard.
- Synthesized the data into 20+ executive-ready insights, led the dialogue with stakeholders and senior team members on how to generate firm-wide impact.

Google Summer of Code *[Google-funded global program for students to contribute to Open Source Software]*

Remote

Software Developer Intern

Apr – Aug 2020

- Refactored major components in the legacy codebase to follow MVC (Model-View-Controller) architecture making the codebase streamlined for future development; Launched an offline version for 100+ students in Peru with limited internet accessibility
- Resolved 71 bugs/feature-requests, added 10+ new blocks in the visual programming interface, integrated USB MIDI input, built an online project sharing platform, developed an oscilloscope - used as a free guitar tuner by 2K+ instructors and learners
- Initiated the development of a fully-programmable Digital Audio Workstation; Used natural language processing (NLP) algorithms, like BERT and TF-IDF, to improve the in-app search precision by +24%

PUBLICATIONS

Leveraging Intra and Inter Modality Relationship for Multimodal Fake News Detection

WWW' 22: The ACM Web Conference | Lyon, France

- Captured the intra-modality relationship via a bottom-up attention approach, making use of R-CNN framework, gathering only the salient patches from images, then finally passing the text through a self-attention module and image patches through multiple CNN layers to obtain intermediate embeddings
- Presented a novel architecture that effectively identifies and suppresses information from weaker modalities and extracts relevant information from strong modality on a per-sample basis using a multiplicative multimodal method; Outperforming existing methodologies by **3.05%** [accuracy] and **4.53%** [F1-score]

ACHIEVEMENTS

Finalist, Mathematical Contest in Modelling (MCM/ICM)

Feb 2023

[Team-IIITD in the annual Modelling competition organized by COMAP]

- Ranked **3rd in India** and within the **top 3% globally** in COMAP's MCM/ICM among 25k teams
- Performed a university-wide NY Times WORDLE survey to model 36 different strategies used by players with varying levels of expertise into an extensive Hierarchical Markov Chain, which can get the correct answer within 4 tries 7/10 times.
- Classified solution words by difficulty using XGBoost trained on historical data to model a negative hypergeometric distribution, also used LIME and SHAP interpretability results to gauge the difficulty attributes of a given word.
- Developed a framework to estimate the distribution of reported results in the NY Times WORDLE game for a future solution word, using a novel technique that combines Dirichlet regression, XGBoost and ARIMA achieving a brier score of <0.1

1st Place, HackPSU

Mar 2021

- International winner among **515** teams @ HackPSU (organized by Penn State University), also won the Nittany Ai Alliance “Ai for Good” Challenge, winning a total sum of **\$1000** in prizes
- Built a browser-extension based personal productivity and mental well-being companion app that tracks browsing activity using the chrome-extension V2 API while keeping your mood in check using the Microsoft Azure Facial Emotion API through the system’s front camera.

ACADEMIC RESEARCH AND PROJECTS

Safe Exploration in Deep Reinforcement Learning

Aug 2022 - Dec 2023

Guide: Prof. Sanjit K Kaul (Course: Reinforcement Learning, IIIT Delhi)

- Addressed the issue of deploying RL Agents on physical environments where violating critical constraints can’t be tolerated, through a novel approach that used a state-based action correction layer along with appropriate reward shaping on top of DDPG/TD3 algorithms
- Showcased the effectiveness of the Agent on Ball-3d and Spaceship Openai gym environments via faster convergence, 10-15% higher cumulative reward than other alternatives, all the while following a zero-violation policy

Detecting Causes of Mental Health Issues in Code-Mixed Social Media Posts

Jan 2023 - May 2023

Guide: Prof. Tavpritesh Sethi (Course: Computing for Medicine, IIIT Delhi)

- Scraped and Augmented a dataset of 12k reddit posts from top Mental Health subreddits, further translated this data into Hindi/English code-mixed using a pretrained multilingual hi-en version of mBART to test the Multilingual robustness of the model
- Trained a Sentence-BERT model on our code-mixed data along with the CAMS dataset to identify underlying causes of mental illness by analysing the user’s social media posts, gaining **+5%** in accuracy over alternative benchmark models

PythonTutor ++

Jan 2023 - May 2023

Guide: Distinguished Prof. Pankaj Jalote (IIIT Delhi)

- Developed an advanced version of pythontutor.com for 200+ students studying Compilers and Intro to Programming at IIIT Delhi, to gain in-depth knowledge about ByteCode and Evaluation-Stack execution at the heart of Python Runtime Systems
- Extracted the Internal Symbols, AST and Evaluation Stack Status from CPython’s runtime at each step of ByteCode execution and displayed it in an interactive NodeJs WebApp for visual learning and debugging

POSITIONS OF RESPONSIBILITY

President: Byld (Software development club @ IIIT Delhi)

Aug 2022 - Feb 2023

- Organized a 2-Day technical hackathon with talks from industry leaders and ex-members on exciting new fields like “Web3 & Blockchain”, “LLMs in Finance” and “Flutter-Development” for a cohort of 200+ Student Attendees and 70 Participating teams
- Redesigned the official [IIITD library portal](#) using Svelte and headless Drupal, decreasing the page load time by 32%; also devised an easy-to-use system for website managers, enabling them to create responsive and interactive web pages using simple md/yaml files
- Facilitated the pairing and outreach of a mentor-mentee cohort of 30+ Freshman/Sophomore students interested in upskilling technically in fields like Web Development, Machine Learning and Open source development; Resulted in 3-Google summer of code, 2-Outreachy and 1-FOSSAsia internship acceptances.

TECHNICAL SKILLS

Programming: R, Python, C/C++

Data Science: SQL, Tableau, MATLAB, Apache Spark, Pandas, NumPy, Seaborn

Machine Learning Frameworks: Scikit-Learn, PyTorch, Keras

ADDITIONAL INFORMATION

Languages: Proficient in English and Hindi

Community Engagement: Participated in trash clean-up activities in community parks, Bengaluru (Goodera, May 2024); Volunteered in local government schools to teach elementary mathematics and serve mid-day meals, Bengaluru (KV DRDO School, July 2023); Helped educating underprivileged kids in basic computer training, Gurugram (Udaan NGO, June 2022)

Online Courses: Neural Networks and Deep Learning (Coursera, 2020); Haskell MOOC (University of Helsinki, 2023)

Interests: Officially ranked 29th (by the SRFi) in U17 Junior Squash; Led the IIITD Squash team to inter-college victory in 2023; completed the “Developing Your Musicianship” Specialization (Berklee, Coursera 2020)